Jacobian-Metric-Energy Geometry

A Differential Theory of Sensitivity, Transformation Cost, Invariants, and Constrained Evolution

Mohsen Dirbaz, PhD

Abstract

We construct a differential framework in which local sensitivity, metric structure, and transformation cost are distinct but coupled mathematical objects. Let a state manifold M carry a task-relative metric g, a transformation map f, and a constraint family defining an admissible tangent cone. The Jacobian describes first-order transport of perturbations; the metric defines comparison or cost of admissible directions; and an energy functional assigns path-dependent burden. We prove that these objects coincide only under additional structural assumptions and therefore must not be conflated. A constrained differential evolution law is obtained by projecting admissible velocities onto the tangent cone under the declared metric. An invariance theorem shows that a differentiable quantity I remains constant along admissible trajectories precisely when its differential annihilates the admissible direction field. We identify the boundary at which differential reasoning fails: discrete transitions, topology changes, and constraint activation can make a local Jacobian insufficient for global transformation.

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1 Introduction

1.1 Three objects often conflated

Differential reasoning about transformation rests on three mathematical objects that are frequently treated as interchangeable:

- the *Jacobian*, which describes how a transformation transports perturbations;
- the *metric*, which defines how directions are compared and costed;
- the *energy functional*, which assigns a burden to a path.

These objects are related but not identical. The Jacobian is a linear map between tangent spaces. The metric is a symmetric positive-definite bilinear form. The energy is a functional on paths. Conflating them produces errors that propagate silently: a transformation may be Jacobian-invertible but metrically expansive, or metrically isometric but energetically costly.

This paper separates the three, proves when they coincide, and characterizes the conditions under which a constrained differential evolution law exists. It then proves an invariance theorem connecting differentiable invariants to the admissible direction field, and identifies the boundary at which differential reasoning fails.

1.2 Relation to companion papers

The companion papers establish the constitutional, decision-theoretic, sufficiency, certification, and decomposition frameworks. The first paper, *Constitutional Geometry of Adaptive Systems*, treats admissibility as a structural property. The second, *From State Space to Decision Space*, formalizes the decision quotient. The third, *The Calculus of Sufficient Distinction*, characterizes sufficiency. The fourth, *Proof-Carrying Optimization*, separates search from certification. The fifth, *Decomposition Without Loss*, characterizes exact decomposition through interface invariants.

The present paper addresses the differential-theoretic question: how do sensitivity, metric, and energy relate, and when do they coincide? This is the question of *Jacobian-metric-energy geometry*, and it is the differential foundation for constrained evolution in the broader program.

1.3 Contributions

1. We define the state manifold, the task-relative metric, the transformation map, and the admissible tangent cone.

2. We define the Jacobian, the metric, and the energy functional as distinct objects and prove a non-conflation theorem: they coincide only under additional structural assumptions.

3. We derive the constrained differential evolution law by projecting admissible velocities onto the tangent cone under the metric.

4. We prove an invariance theorem: a differentiable quantity I is constant along admissible trajectories if and only if its differential annihilates the admissible direction field.

5. We identify the boundary of differential reasoning: discrete transitions, topology changes, and constraint activation.

6. We provide constructive counterexamples for each failure mode and falsification strategies for every substantive claim.

1.4 Falsification strategy

Every substantive claim is falsifiable by a structural counterexample. The non-conflation theorem fails if there exist a manifold, a transformation, a metric, and an energy functional for which the three objects coincide without the stated structural assumptions. The invariance theorem fails if there exists a differentiable quantity that is constant along admissible trajectories but whose differential does not annihilate the admissible direction field, or vice versa. The boundary claims fail if a discrete transition, a topology change, or a constraint activation can be represented locally by a Jacobian.

2 Manifold Structure

2.1 State manifold

Definition 2.1 (State manifold)**.** *A* state manifold *is a smooth finite-dimensional manifold M without boundary.*

The state manifold is the differentiable state space. Its points are states; its tangent spaces are directions of infinitesimal change.

Definition 2.2 (Tangent bundle)**.** *The* tangent bundle *of M is $TM = \bigsqcup_{p \in M} T_pM$. Elements of T_pM are* tangent vectors *at p.*

Definition 2.3 (Vector field)**.** *A* vector field *on M is a smooth section $V : M \to TM$ with $V(p) \in T_pM$.*

A vector field assigns a direction to every state. It is the infinitesimal generator of a flow.

2.2 Transformation map

Definition 2.4 (Transformation map)**.** *A* transformation map *is a smooth map $f : M \to M$.*

A transformation map sends a state to a successor state. Its differential is the Jacobian.

Definition 2.5 (Jacobian)**.** *The* Jacobian *of f at $p \in M$ is the linear map*

$$Df_p : T_pM \to T_{f(p)}M.$$

When M is a vector space with coordinates, Df_p is represented by the matrix of partial derivatives.

The Jacobian describes first-order transport of perturbations. It is a linear map between tangent spaces at p and $f(p)$.

Proposition 2.6 (Chain rule)**.** *If $f, h : M \to M$ are smooth, then*

$$D(h \circ f)_p = Dh_{f(p)} \circ Df_p.$$

Proof. Standard differential geometry. □

2.3 Curves and flows

Definition 2.7 (Curve)**.** *A* curve *on M is a smooth map $\gamma : [0, T] \to M$.*

Definition 2.8 (Velocity)**.** *The* velocity *of a curve γ at t is*

$$\dot{\gamma}(t) = \frac{d\gamma}{dt}(t) \in T_{\gamma(t)}M.$$

Definition 2.9 (Flow)**.** *A* flow *on M is a smooth map $\phi : \mathbb{R} \times M \to M$ satisfying:*

1. *$\phi(0, p) = p$ for all $p \in M$;*
2. *$\phi(s + t, p) = \phi(s, \phi(t, p))$ for all $s, t \in \mathbb{R}$, $p \in M$.*

The infinitesimal generator *of the flow is the vector field*

$$V(p) = \frac{\partial \phi}{\partial t}(0, p).$$

Proposition 2.10. *Every smooth vector field with compact support generates a unique local flow.*

Proof. Standard existence and uniqueness theory for ordinary differential equations (Picard–Lindelöf). □

3 The Three Objects

3.1 The Jacobian

The Jacobian $Df_p : T_pM \to T_{f(p)}M$ is a linear map between distinct tangent spaces. It describes how a perturbation $\delta p \in T_pM$ is transported to a perturbation $Df_p(\delta p) \in T_{f(p)}M$.

Proposition 3.1 (Jacobian composition)**.** *For a curve γ and a transformation f, the velocity of $f \circ \gamma$ is*

$$\frac{d}{dt}(f \circ \gamma)(t) = Df_{\gamma(t)}(\dot{\gamma}(t)).$$

Proof. Chain rule. □

The Jacobian is the *transport* of directions. It does not define a comparison between directions at the same point.

3.2 The metric

Definition 3.2 (Metric)**.** *A* metric *on M is a smooth section g of $T^*M \otimes T^*M$ such that for each $p \in M$, the bilinear form*

$$g_p : T_pM \times T_pM \to \mathbb{R}$$

is symmetric and positive definite.

Definition 3.3 (Length of a tangent vector)**.** *The* length *of $v \in T_pM$ is*

$$\|v\|_g = \sqrt{g_p(v, v)}.$$

Definition 3.4 (Length of a curve)**.** *The* length *of a curve $\gamma : [0, T] \to M$ is*

$$L_g[\gamma] = \int_0^T \|\dot{\gamma}(t)\|_g \, dt.$$

The metric defines comparison and cost of directions. It measures the size of perturbations and the length of paths.

Proposition 3.5 (Metric as a bilinear form)**.** *For each $p \in M$ and $v, w \in T_pM$, the quantity $g_p(v, w)$ is a symmetric bilinear form. When $v = w$, it is the squared length of v.*

Proof. Symmetry and bilinearity are properties of the metric. Positive definiteness ensures $g_p(v, v) \geq 0$ with equality iff $v = 0$. □

The metric is the *comparison* of directions at the same point. It does not, by itself, transport directions.

3.3 The energy functional

Definition 3.6 (Energy functional)**.** *An* energy functional *is a functional*

$$E : \{\textit{curves on } M\} \to \mathbb{R}$$

that assigns a real number to each curve.

Definition 3.7 (Path energy)**.** *The* path energy *of a curve γ is*

$$E[\gamma] = \int_0^T \mathcal{L}(\gamma(t), \dot{\gamma}(t))\, dt$$

where $\mathcal{L} : TM \to \mathbb{R}$ is a smooth Lagrangian*.*

The energy is *path-dependent*: it depends on the entire curve, not merely on its endpoints or its length.

Definition 3.8 (Energy from the metric)**.** *If $\mathcal{L}(p, v) = \frac{1}{2} g_p(v, v)$, then the path energy is*

$$E_g[\gamma] = \frac{1}{2} \int_0^T g_{\gamma(t)}(\dot{\gamma}(t), \dot{\gamma}(t))\, dt = \frac{1}{2} \int_0^T \|\dot{\gamma}(t)\|_g^2\, dt.$$

This is the special case in which the energy is determined by the metric. It is the *kinetic energy* of the path.

Proposition 3.9 (Energy is not length)**.** *In general, $E_g[\gamma] \neq L_g[\gamma]$. Equality holds if and only if $\|\dot{\gamma}(t)\|_g$ is constant along γ and normalized so that $E_g[\gamma] = \frac{1}{2} L_g[\gamma]^2$.*

Proof. By Cauchy–Schwarz,

$$L_g[\gamma]^2 = \left(\int_0^T \|\dot{\gamma}\|_g\, dt \right)^2 \leq T \int_0^T \|\dot{\gamma}\|_g^2\, dt = 2T E_g[\gamma].$$

Equality holds iff $\|\dot{\gamma}\|_g$ is constant. Then $E_g = \frac{1}{2} T \|\dot{\gamma}\|_g^2 = \frac{1}{2T} L_g^2$. With normalization $T = 1$ or $\|\dot{\gamma}\|_g = 1$, the relation holds. □

Remark 3.10. *The energy and the length are distinct functionals. They coincide only in the special case of constant speed. This is the first instance of the non-conflation theorem.*

4 The Non-Conflation Theorem

4.1 Statement

Theorem 4.1 (Non-conflation)**.** *The Jacobian, the metric, and the energy functional are distinct mathematical objects. They coincide only under additional structural assumptions.*

Proof. We exhibit three scenarios.

Scenario 1: Jacobian without metric. Let $M = \mathbb{R}^n$ with the standard smooth structure but no metric. A transformation $f : M \to M$ has a Jacobian Df_p defined by partial derivatives, but no metric is available for comparison. The Jacobian exists, but the metric does not.

Scenario 2: Metric without Jacobian. Let M be a Riemannian manifold with metric g. The metric exists, but without a specified transformation f, there is no Jacobian. The metric is a structural property of the manifold; the Jacobian requires a transformation.

Scenario 3: Jacobian and metric without energy. Let $M = \mathbb{R}^n$ with Euclidean metric g and a linear transformation $f(x) = Ax$. The Jacobian is A; the metric is the identity. The energy is determined by a Lagrangian, which is not specified by the Jacobian or the metric alone. Different Lagrangians give different energies on the same (M, g, f). □

Remark 4.2. *The theorem is the formal content of the abstract's claim that the three objects are distinct. They are coupled—the metric can be transported by the Jacobian, and the energy can be defined using the metric—but they are not identical.*

4.2 Structural conditions for coincidence

When do the three objects coincide?

Definition 4.3 (Isometric transformation)**.** *A transformation $f : M \to M$ is* isometric *if*

$$g_{f(p)}(Df_p(v), Df_p(w)) = g_p(v, w)$$

for all $p \in M$ and $v, w \in T_pM$.

An isometry preserves the metric. Its Jacobian is a linear isometry between tangent spaces.

Theorem 4.4 (Coincidence under isometry)**.** *If f is an isometry and the Lagrangian is $\mathcal{L}(p, v) = \frac{1}{2}g_p(v, v)$, then the Jacobian transports the metric: for any curve γ,*

$$L_g[f \circ \gamma] = L_g[\gamma], \quad E_g[f \circ \gamma] = E_g[\gamma].$$

The Jacobian, metric, and energy are consistent: the Jacobian preserves the metric, and the energy is invariant under the Jacobian.

Proof. For the length:

$$\|\frac{d}{dt}(f \circ \gamma)(t)\|_g = \|Df_{\gamma(t)}(\dot{\gamma}(t))\|_g = \|\dot{\gamma}(t)\|_g$$

by the isometry property. Integrating gives $L_g[f \circ \gamma] = L_g[\gamma]$.

For the energy:

$$g_{f(\gamma(t))}(Df_{\gamma(t)}(\dot{\gamma}), Df_{\gamma(t)}(\dot{\gamma})) = g_{\gamma(t)}(\dot{\gamma}, \dot{\gamma}).$$

Integrating gives $E_g[f \circ \gamma] = E_g[\gamma]$. □

Corollary 4.5. *The three objects coincide if and only if the transformation is an isometry and the energy is the kinetic energy of the metric.*

Remark 4.6. *Coincidence is a strong structural condition. It holds for isometries (rigid motions, symmetries) but fails for general transformations. This is why the three objects must be distinguished in general.*

4.3 The transport of the metric

Definition 4.7 (Pullback metric)**.** *The* pullback *of the metric g under f is the section f^*g defined by*

$$(f^*g)_p(v, w) = g_{f(p)}(Df_p(v), Df_p(w)).$$

The pullback metric measures directions at p using the metric at $f(p)$, transported through the Jacobian.

Proposition 4.8. *f is an isometry if and only if $f^*g = g$.*

Proof. Immediate from the definitions. □

Definition 4.9 (Metric distortion)**.** *The* distortion *of f at p is*

$$\text{dist}(f, p) = \sup_{v \neq 0} \frac{\|Df_p(v)\|_g}{\|v\|_g}.$$

The distortion measures how much the transformation stretches or shrinks directions. It is 1 for isometries and greater than 1 for expansive transformations.

Proposition 4.10 (Distortion and the metric)**.** *The distortion is finite and positive for every smooth f with Df_p invertible.*

Proof. The distortion is the operator norm of Df_p with respect to g. It is finite because T_pM is finite-dimensional and Df_p is bounded. It is positive because Df_p is invertible. □

5 Constrained Evolution

5.1 The admissible tangent cone

Definition 5.1 (Constraint family)**.** *A* constraint family *on M is a collection of smooth functions*

$$\mathcal{C} = \{c_\alpha : M \to \mathbb{R}\}_{\alpha \in A}$$

called constraint functions*.*

Definition 5.2 (Constraint set)**.** *The* constraint set *is*

$$K = \{p \in M : c_\alpha(p) \geq 0 \text{ for all } \alpha \in A\}.$$

Definition 5.3 (Active constraint)**.** *A constraint c_α is* active *at $p \in K$ if $c_\alpha(p) = 0$, and* inactive *if $c_\alpha(p) > 0$.*

Definition 5.4 (Admissible tangent cone)**.** *The* admissible tangent cone *at $p \in K$ is*

$$T_pK = \{v \in T_pM : dc_\alpha|_p(v) \geq 0 \text{ for all active } \alpha\}.$$

The admissible tangent cone is the set of directions that do not immediately violate the constraints. Directions in T_pK are *admissible velocities*.

Proposition 5.5. *T_pK is a closed convex cone in T_pM.*

Proof. T_pK is the intersection of half-spaces $\{v : dc_\alpha|_p(v) \geq 0\}$ over active α. Each half-space is closed and convex. Their intersection is closed and convex. It is a cone because it is closed under positive scaling. □

5.2 The constrained evolution law

Definition 5.6 (Constrained evolution law)**.** *Given a desired velocity field $V : M \to TM$ and a metric g, the* constrained evolution law *is*

$$V^K(p) = \Pi^g_{T_pK}(V(p))$$

where $\Pi^g_{T_pK}$ is the orthogonal projection onto T_pK under the metric g.

The constrained evolution law projects the desired velocity onto the admissible tangent cone. It is the closest admissible velocity to the desired velocity, measured by g.

Proposition 5.7 (Variational characterization)**.** *$V^K(p)$ is the unique solution of*

$$\min_{w \in T_pK} \|V(p) - w\|_g^2.$$

Proof. The projection onto a closed convex set in a Hilbert space is the unique nearest point. □

Definition 5.8 (Flow)**.** *The* constrained flow *of the evolution law is the flow of the vector field V^K, restricted to K.*

Proposition 5.9 (Constraint preservation)**.** *The constrained flow preserves K: if $p \in K$ and $\phi_t^K(p)$ is the constrained flow, then $\phi_t^K(p) \in K$ for all $t \geq 0$ in the flow domain.*

Proof. The evolution law satisfies $dc_\alpha(V^K(p)) \geq 0$ for all active α. By Nagumo's theorem, the flow preserves K. □

Remark 5.10. *The constrained evolution law is the differential form of constitutional transition. Where Article 1 defined constitutional transitions as partial maps on the admissible set, the present paper defines them as projections of velocities onto the admissible tangent cone.*

5.3 The projection formula

When constraints are given by smooth functions with linearly independent gradients, the projection has a closed form.

Definition 5.11 (Lagrange multiplier form)**.** *Let $\{\alpha_1, \ldots, \alpha_m\}$ be the active constraints at p with linearly independent gradients. Define the matrix*

$$G_{ij} = g_p(\nabla c_{\alpha_i}, \nabla c_{\alpha_j}).$$

The Lagrange multipliers *are $\lambda = G^{-1}\mu$ where*

$$\mu_i = g_p(\nabla c_{\alpha_i}, V(p)).$$

Proposition 5.12. *The projected velocity is*

$$V^K(p) = V(p) - \sum_i \lambda_i \nabla c_{\alpha_i}.$$

Proof. The projection onto a convex cone defined by linear inequalities with linearly independent gradients satisfies the KKT conditions:

$$V^K = V - \sum_i \lambda_i \nabla c_{\alpha_i}, \quad \lambda_i \geq 0, \quad g_p(V^K, \nabla c_{\alpha_i}) = 0 \text{ for all } i.$$

Solving for λ_i gives the stated formula. □

Remark 5.13. *The projection formula is the differential analogue of the constitutional transition definition. The Lagrange multipliers are the "prices" of the constraints.*

6 The Invariance Theorem

6.1 Differentiable invariants

Definition 6.1 (Differentiable invariant)**.** *A smooth function $I : M \to \mathbb{R}$ is a* differentiable invariant *along a vector field V if*

$$dI|_p(V(p)) = 0 \quad \text{for all } p \in M.$$

Definition 6.2 (Admissible direction field)**.** *The* admissible direction field *is the assignment* $p \mapsto T_pK$ *of the admissible tangent cone at each point.*

Definition 6.3 (Invariance along admissible trajectories)**.** *A smooth function* $I : M \to \mathbb{R}$ *is* invariant along admissible trajectories *if for every admissible curve* γ *(i.e.,* $\dot{\gamma}(t) \in T_{\gamma(t)}K$ *for all* t*),*

$$\frac{d}{dt} I(\gamma(t)) = 0.$$

6.2 Statement and proof

Theorem 6.4 (Invariance)**.** *A smooth function* $I : M \to \mathbb{R}$ *is invariant along admissible trajectories if and only if*

$$dI|_p(v) = 0 \quad \text{for all } p \in K \text{ and all } v \in T_pK.$$

That is, the differential of I *annihilates the admissible direction field.*

Proof. ($\Leftarrow$) Let γ be an admissible curve. Then

$$\frac{d}{dt} I(\gamma(t)) = dI|_{\gamma(t)}(\dot{\gamma}(t)) = 0$$

because $\dot{\gamma}(t) \in T_{\gamma(t)}K$.

($\Rightarrow$) Suppose I is invariant along admissible trajectories. Let $p \in K$ and $v \in T_pK$. Let $\gamma : [0, \epsilon) \to K$ be an admissible curve with $\gamma(0) = p$ and $\dot{\gamma}(0) = v$. Such a curve exists by the definition of the tangent cone. Then

$$0 = \frac{d}{dt}\bigg|_{t=0} I(\gamma(t)) = dI|_p(v).$$

□

Corollary 6.5 (First integrals of the constrained flow)**.** *If* I *satisfies the invariance condition, then* I *is a first integral of the constrained flow:*

$$I(\phi_t^K(p)) = I(p)$$

for all $p \in K$ *and all* t *in the flow domain.*

Proof. The flow $\phi_t^K(p)$ is an admissible curve. Apply Theorem 6.4. □

Remark 6.6. *The theorem is the formal content of the abstract's claim that a differentiable quantity* I *remains constant along admissible trajectories precisely when its differential annihilates the admissible direction field. The condition is exact: invariance is equivalent to the annihilation.*

6.3 Consequences

Corollary 6.7 (Invariants of constrained evolution)**.** *The invariants of the constrained flow are exactly the functions whose differentials annihilate* T_pK *at every* $p \in K$*.*

Proof. Theorem 6.4. □

Proposition 6.8 (Active constraints as invariants)**.** *If* c_α *is a constraint with* $c_\alpha(p) = 0$ *for all* p *in an open subset of* K*, then* c_α *is an invariant on that subset.*

Proof. For p in the subset, c_α is active and $dc_\alpha|_p(v) = 0$ for all $v \in T_pK$ (since the constraint is satisfied on the boundary). By Theorem 6.4, c_α is invariant. □

Remark 6.9. *Active constraints become invariants when the system evolves on the constraint boundary. This is the geometric origin of conservation laws in constrained systems.*

7 The Boundary of Differential Reasoning

7.1 Three failure modes

The differential framework assumes smoothness. Three classes of phenomena escape it.

7.2 Discrete transitions

Definition 7.1 (Discrete transition)**.** *A* discrete transition *is a map $\delta : M \to M$ that is not smooth at any point.*

Proposition 7.2. *A discrete transition has no Jacobian at any point.*

Proof. The Jacobian $D\delta_p$ is defined only if δ is differentiable at p. A discrete transition is not differentiable. □

Example 7.3. *Let $M = \mathbb{R}$ and $\delta(x) = \lfloor x \rfloor$. The Jacobian does not exist at integer points.*

Remark 7.4. *Discrete transitions cannot be represented locally by a Jacobian. They require separate jump semantics. This is the first boundary of differential reasoning.*

7.3 Topology changes

Definition 7.5 (Topology change)**.** *A* topology change *is a transformation that changes the topological type of the state space, e.g., the number of connected components, the genus of a surface, or the homotopy type.*

Proposition 7.6. *A topology change cannot be represented by a smooth map between manifolds of the same dimension.*

Proof. A smooth map between manifolds of the same dimension is a local diffeomorphism on its regular points. It preserves the local topology. A topology change requires either a change in dimension or a singularity. □

Example 7.7. *A sphere cannot be smoothly deformed into a torus.*

Remark 7.8. *Topology changes require a sequence of structural maps between state strata. This is the second boundary of differential reasoning.*

7.4 Constraint activation

Definition 7.9 (Constraint activation)**.** *A constraint c_α* activates *at $p \in K$ if $c_\alpha(p) = 0$ and there exists a neighborhood U of p such that c_α is inactive on $U \cap K \setminus \{p\}$.*

Proposition 7.10. *At a constraint activation point, the projection onto T_pK is not differentiable with respect to p.*

Proof. The projection depends on the set of active constraints, which changes discontinuously when a constraint activates. The change in the active set is a discontinuous change in the projection map. □

Example 7.11. *Let $M = \mathbb{R}^2$ and $K = \{(x, y) : y \geq 0\}$. Let $V(x, y) = (0, -1)$. The constrained velocity is $V^K = (0, 0)$ on the boundary $y = 0$ and $V^K = (0, -1)$ in the interior. The projection is discontinuous at the boundary.*

Remark 7.12. *Constraint activation requires a hybrid or switched system. This is the third boundary of differential reasoning. It is the differential analogue of the constitutional transition: the system changes behavior when it hits the constraint.*

7.5 The boundary theorem

Theorem 7.13 (Boundary of differential reasoning)**.** *The differential framework applies to a transformation if and only if:*

1. *the transformation is smooth;*
2. *the topology of the state space is preserved;*
3. *the active constraint set is locally constant.*

Outside these conditions, differential reasoning is insufficient and must be supplemented by jump semantics, structural maps, or hybrid dynamics.

Proof. The Jacobian requires smoothness. The flow requires a fixed manifold. The projection requires a constant active constraint set. Violating any condition invalidates the corresponding differential tool. □

Remark 7.14. *The theorem is the formal content of the abstract's claim that discrete transitions, topology changes, and constraint activation can make a local Jacobian insufficient for global transformation. The three conditions are the exact boundary.*

8 Falsifiers and Structural Counterexamples

8.1 Falsifier for the non-conflation theorem

A counterexample to Theorem 4.1 would be a manifold, a transformation, a metric, and an energy functional for which the three objects coincide without the structural conditions of Theorem 4.4. This would require a non-isometric transformation to preserve the metric, which contradicts the definition of an isometry.

8.2 Falsifier for the coincidence theorem

A counterexample to Theorem 4.4 would be an isometry f and a curve γ with $E_g[f \circ \gamma] \neq E_g[\gamma]$. This is impossible by the proof: the isometry property is exactly the condition needed for the energy to be invariant.

8.3 Falsifier for the constrained evolution

A counterexample to the projection formula would be a constraint family with linearly independent gradients for which the projection does not satisfy the KKT conditions. This would violate the standard theory of projection onto convex cones.

8.4 Falsifier for the invariance theorem

A counterexample to Theorem 6.4 would be a smooth function I that is invariant along admissible trajectories but whose differential does not annihilate T_pK at some point. This would require an admissible direction v with $dI|_p(v) \neq 0$, which contradicts invariance by the curve construction in the proof.

8.5 Falsifier for the boundary theorem

A counterexample to Theorem 7.13 would be a discrete transition, topology change, or constraint activation that can be represented by a Jacobian. This would violate the differentiability, homeomorphism, or local constancy conditions.

8.6 How to test the theory

1. Specify M, the metric g, the transformation f, and the constraint family $\mathcal{C}$.
2. Compute the Jacobian Df_p and verify that it is a linear map between tangent spaces.
3. Compute the pullback metric f^*g and check whether f is an isometry.
4. For a non-isometric f, compute the distortion.
5. Compute the admissible tangent cone T_pK at points of interest.
6. Project a desired velocity onto T_pK and verify the KKT conditions.
7. Identify invariants by checking the annihilation condition.
8. Check the boundary conditions: smoothness, topology preservation, constraint activation.

   Any failure is a falsifier.

9 Discussion

9.1 The three objects

The Jacobian, the metric, and the energy functional are distinct. The Jacobian transports perturbations. The metric compares directions. The energy assigns path-dependent burden. They coincide only for isometries with kinetic energy. Conflating them produces errors: a transformation may be Jacobian-invertible but metrically expansive, or metrically isometric but energetically costly.

9.2 Constrained evolution

The constrained evolution law projects desired velocities onto the admissible tangent cone. The projection is well-defined, unique, and preserves the constraint set. The projection formula uses Lagrange multipliers to account for active constraints.

9.3 Invariance

The invariance theorem gives an exact characterization: a smooth function is invariant along admissible trajectories if and only if its differential annihilates the admissible direction field. This is the differential form of the invariant-preservation theorem of Article 1.

9.4 Boundary

Discrete transitions, topology changes, and constraint activation mark the boundary of differential reasoning. Outside this boundary, differential tools must be supplemented by jump semantics, structural maps, or hybrid dynamics. This is the differential form of the transition-composition theorem of Article 1.

9.5 Relation to companion papers

The first paper established admissibility as structural. The second established the decision quotient. The third established sufficiency. The fourth established certification. The fifth established decomposition. The present paper establishes the differential geometry of constrained evolution.

Together the six papers provide a complete account of governed computation: admissibility selects the feasible set, the decision quotient selects the decision space, sufficiency determines whether a representation preserves decision-relevant distinctions, certification establishes when a candidate is optimal, decomposition determines when the problem factors, and differential geometry determines how constrained systems evolve.

9.6 Limitations

The framework does not address:

- How to construct the metric in practice.
- How to choose the Lagrangian.
- How to handle non-smooth constraints.
- How to compute the constrained flow numerically.

These are domain-specific questions.

9.7 Domain neutrality

The framework applies to any system with:

- a smooth state manifold,
- a transformation map,
- a metric,
- an energy functional,
- a constraint family.

This includes mechanical systems, thermodynamic systems, quantum systems, control systems, and any system whose evolution is governed by differential equations on a constrained manifold.

10 Conclusion

We have constructed a differential framework in which local sensitivity, metric structure, and transformation cost are distinct but coupled. The main results are:

1. The non-conflation theorem: the Jacobian, the metric, and the energy functional are distinct and coincide only for isometries with kinetic energy.
2. The constrained evolution law: admissible velocities are projections onto the admissible tangent cone under the metric.

3. The invariance theorem: a differentiable quantity is invariant along admissible trajectories if and only if its differential annihilates the admissible direction field.

4. The boundary theorem: differential reasoning fails at discrete transitions, topology changes, and constraint activation.

The framework is falsifiable by structural counterexamples and is domain-neutral. Its purpose is to make the distinctions between sensitivity, metric, and cost precise, and to characterize the exact conditions under which constrained differential evolution is possible.

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